WACV #*****

Reading Unseen Scripts: Predictable Zero-Real-Image Transfer for Brahmic Scene Text Recognition

Anonymous WACV Algorithms Track submission

Paper ID *****

Abstract

Scene-text recognition (STR) is effectively solved for Latin script yet fails for most of the world's writing systems, and collecting labeled real images for every script is not a scalable answer. We ask a stronger question: can a recognizer read a script for which no real labeled image exists at all — and can its accuracy be predicted before training? We construct a shared grapheme pivot space spanning nine Brahmic scripts and fine-tune a compact open vision–language model in it under a leave-one-script-out protocol: the held-out script is seen only as rendered synthetic images, never as a real photograph. On real scene-text benchmarks the held-out scripts reach 9.16–30.09% word recognition (character accuracy up to 57.77%), against source-only baselines of 0.0–5.46%. Replacing the grapheme pivot with the stock BPE tokenizer degrades word recognition on all nine scripts (1.08×–3.91×; exact sign test $p = 0.0039$), on three of them despite equal-or-better character accuracy — the pivot, not fine-tuning alone, converts character evidence into word reads. The campaign was run under a frozen preregistration: our primary hypothesis — that tokenizer fertility predicts which scripts transfer — was falsified (Spearman −0.80) and is reported as such; the surviving two-factor account (synthetic-exposure budget and pivot-space coverage) was itself tested prospectively, with point predictions and error bands committed to a version-controlled archive before each remaining run and scored as hits and misses. We release the pivot construction, rendering pipeline, and the zero-real-image evaluation protocol.

1. Introduction

Modern scene-text recognizers exceed 90% word accuracy on Latin benchmarks, yet most of the world's writing systems remain unreadable to them. Recent audits are stark: across more than a hundred Unicode scripts, even frontier vision–language models (VLMs) read fewer than thirty, and on unfamiliar scripts they emit garbled or substituted output [14]. For the Brahmic family — nine major scripts used by over a billion people — real-image training data exists only because of recent, labor-intensive benchmark efforts [8, 19], and dozens of related scripts (Sinhala, Khmer, Myanmar, Tibetan, ...) have little or none. Collecting and annotating real photographs script-by-script is not a scalable path to coverage.

This paper asks a stronger question than "how well can script X be read with its training set": can a recognizer read a script for which no real labeled image exists at all — and can we predict, before training, how well? We answer both, for scene text, at the granularity of an entire writing system.

Our approach rests on a structural observation about Brahmic scripts: they descend from a common ancestor and share an organizational scheme — consonant bases, vowel signs, and conjunct formation via a virama — even though their glyphs and Unicode blocks are disjoint. We exploit this by constructing a shared grapheme pivot space: every script's text is decomposed into grapheme clusters and mapped by structural correspondence into a common output vocabulary, so that nine visually unrelated scripts write into one label space. A compact open VLM (Florence-2, 0.23B parameters [24]) is fine-tuned in this space on real images of source scripts plus rendered synthetic images of the target script — never a real target photograph.

We evaluate with a leave-one-script-out (LOSO) protocol over all nine scripts of a public real-image benchmark [8]. For each held-out script we train two models: a source-only model (Rung A), and a source+synthetic-target model (Rung B); both are evaluated on the held-out script's real test images. Transfer is substantial and universal: every script improves from a 0.0–5.46% baseline to 9.16–30.09% word recognition rate (WRR), with character accuracy up to 57.77% — using zero real target images. An ablation that swaps the grapheme pivot for the stock BPE tokenizer, all else identical, degrades transfer on all nine scripts (1.08×–3.91×; exact sign test $p = 0.0039$); strikingly, on Telugu the BPE model matches the pivot model's character accuracy (55.1% vs. 54.43%) while halving word accuracy — the pivot is what converts character evidence into whole-word reads.

The second contribution is epistemic, and to our knowledge unprecedented in text recognition: the campaign was run under a frozen, version-controlled preregistration with an append-only amendment log, and its predictions were filed prospectively — committed and pushed before the runs they predict. This cuts both ways, and we report both edges. Our preregistered primary hypothesis — that tokenizer fertility, the property that governs the supervised grapheme-vs-BPE trade-off (Sec. 3), also predicts which scripts transfer — was falsified (Spearman −0.80 across nine scripts). The account that survived is a two-factor one: synthetic-exposure budget dominates, and within a fixed budget, pivot-space coverage orders transfer. Its out-of-sample record is auditable: a point forecast of ∼15 WRR for Kannada (realized 15.42); a final-script forecast for Gurmukhi of 16.2 WRR with $\pm 2\sigma$ band [8.2, 24.1], filed and pushed 3.7 hours before training began (realized 22.16: outer band hit, point undercalled — reported as the miss it is).

Our contributions:

1. Zero-real-image cross-script transfer. The first demonstration, to our knowledge, that real scene text of an entirely unseen script — disjoint Unicode block, disjoint glyph inventory — can be read by transferring through a designed structural pivot space, evaluated LOSO across nine Brahmic scripts (9.16–30.09% WRR from $\leq 5.46\%$ baselines).
2. The pivot is the mechanism. A preregistered stock-BPE ablation under identical data, seed, and recipe degrades WRR on all nine scripts (1.08×–3.91×; exact sign test $p = 0.0039$), isolating output-space structure — not synthetic fine-tuning per se — as the transfer mechanism; exploratorily, the size of the penalty tracks the script's fertility (Sec. 5.1).
3. Predictability, tested prospectively. A two-factor model (synthetic budget + pivot coverage) with committed-before-run forecasts, scored hits and misses, a falsified preregistered primary reported in full, and a dose–response synthetic-scaling study (Sec. 4).
4. Artifacts. The pivot construction, rendering and verification pipeline, evaluation protocol, and the preregistration/prediction chain itself, released as a reusable template for prospective empirical ML.

We do not claim per-script state of the art: supervised recognizers trained on real labeled target images reach ∼73% on this benchmark [8] and sit on a different axis — they presuppose the data whose absence defines our problem. The relevant comparisons, which we report, are source-only baselines, the BPE ablation, raw and frontier VLMs on the same test sets, and our own prospective forecasts.

2. Related Work

Multilingual STR and Indic benchmarks. STR is mature for Latin script [3], but coverage collapses elsewhere. IndicSTR12 [19] and the Bharat Scene Text Dataset [8] established real-image benchmarks for Indian languages; on the latter, PARSeq reaches ∼47% average WRR trained on synthetic data only and ∼73% after fine-tuning on real labeled images — numbers that presuppose labeled target data. GlotOCR [14] quantifies the wider problem: frontier VLMs read fewer than thirty of 100+ scripts. Synthetic-data engines for STR remain Latin-centric [25], and fine-tuning VLMs on synthetic renderings has been shown for a single low-resource script with real-data validation [6]. Our work targets the regime where no real labeled target image exists, and asks not only whether such scripts can be read but how well, predicted in advance.

Tokenization granularity for complex scripts. MGP-STR [7] fuses character, BPE, and WordPiece streams within one Latin recognizer; grapheme-level tokenization beats byte-BPE for Bengali handwriting [11]. In the LLM literature, fertility's value as a tokenizer metric is contested [20]. Our preliminary study (Sec. 3) makes the supervised grapheme-vs-BPE choice predictable across nine scripts (polarity 8/9, leave-one-script-out 90.9%), and this paper then tests — and refutes — the natural conjecture that the same property predicts transfer.

Unseen units vs. unseen scripts. A substantial line reads unseen characters within a known script by decomposition into shared sub-units: stroke- and radical-based zero-shot Chinese recognition [4, 26, 27], and open-set recognition of novel characters [16–18]. Zero-shot detection of unseen scripts localizes but does not read [15]. Byte-level pivots let ASR cover languages with unseen scripts in speech [5], and task-arithmetic analogies achieve zero-real-data synthetic-to-real HTR — for five Latin-script languages, with unseen writing systems explicitly left open [10]. To our knowledge, no

WACV #*****

prior system reads real scene text of an entirely unseen script — different Unicode block, different glyphs — from zero real target images, let alone with accuracy forecast beforehand.

What predicts cross-lingual transfer. In NLP, transfer tracks lexical and subword overlap rather than language relatedness [9, 22]. For STR specifically, Ref. [1] concludes that source dataset size and visual similarity, not linguistic typology, drive cross-lingual transfer. Our findings sharpen rather than contradict this, in a regime that work did not study (zero real target images): (i) a typology-adjacent structural property — fertility — indeed fails as a transfer predictor (-0.80), as a data-centric view expects; (ii) synthetic target exposure dominates — a target-side data-size effect; (iii) within a fixed budget, coverage of the shared grapheme space orders transfer ($+0.61$ among equal-budget scripts) — an OCR instantiation of the subword-overlap principle in a designed output space; and (iv) a frozen-encoder visual-similarity control predicts nothing in our data (-0.14). Because LOSO balances source data by construction, the confound identified by [1] is controlled by design.

Prediction and preregistration in empirical ML. Scaling laws make supervised STR predictable in model and data size [23]; preregistration and negative-result reporting are the reproducibility literature's recommended remedies for HARKing and cherry-picking [12, 13], yet full-study adoption is rare. We run the entire campaign under a frozen preregistration with an append-only amendment log and commit quantitative forecasts before the runs they predict; we offer the protocol itself as a contribution.

3. Method

3.1. A shared grapheme pivot space for Brahmic scripts

Brahmic scripts encode the same phonological skeleton — consonant bases, dependent vowel signs, and conjunct formation through a virama — and Unicode preserves this: corresponding letters sit at the same offset within each script's block. We exploit both layers.

Structural pivot. A mapping π sends any Brahmic codepoint to a pivot codepoint by its within-block offset ($\pi(c) = \text{0x0900} + (c - \text{base}(c))$); shared punctuation that is not offset-safe is excluded by construction. Because π is a per-script bijective shift, it is exactly invertible: a hypothesis decoded in pivot space is read back into the target script deterministically. Text from nine scripts thereby writes into one label space in which structurally corresponding syllables become identical strings.

Grapheme-cluster vocabulary. Within pivot space we segment text into grapheme clusters (consonant–virama conjuncts kept whole, dependent signs attached) and inject the resulting cluster inventory (1,624 types for the source pool) as dedicated tokens into the recognizer's tokenizer. The alternative — letting the stock byte-BPE tokenizer segment pivot-space text — is the ablation of Sec. 4: it sees the same pivot strings but fragments clusters into sub-syllabic pieces. Fertility (mean clusters per word) governs when grapheme tokenization beats BPE in supervised training: in a 12-run balanced-budget study across the same nine scripts, fertility predicts the winning branch with 8/9 polarity and 90.9% leave-one-script-out direction accuracy ($R^2 = 0.678$). That result motivates both the pivot design and the (ultimately falsified) primary hypothesis of Sec. 5.

3.2. Zero-real-image protocol

For each held-out script S of the nine in the benchmark [8]:

Sources. All benchmark languages whose script is not S (languages sharing S, e.g. Assamese for Bengali script, are excluded — S is genuinely unseen). Each source language contributes exactly 700 train / 90 val real images (seed 42), giving balanced source exposure by construction; source-side data size is therefore controlled, unlike prior cross-lingual STR studies [1].

Synthetic target. 3,240 images per target language: benchmark-vocabulary words rendered in fontconfig-verified fonts (multiple faces per script, libraqm shaping), each render checked non-blank; the rendered word enters training as its pivot transcription. No real photograph of S is ever seen. Scripts written by two source languages receive two languages' worth of renderings (2× budget) — a covariate disclosed at preregistration time that becomes central in Sec. 5.

Rungs. Rung A trains on sources only; Rung B adds the synthetic target set; Rung B_{bpe} repeats Rung B with the grapheme-cluster injection removed (stock tokenizer), identical data, seed, and recipe. All rungs fine-tune Florence-2 (0.23B) [24] with LoRA ($r = 16$, $\alpha = 32$, dropout 0.05), lr 10^{-4}, batch size 4, 15 epochs, validation-selected checkpoint, beam-search decoding at test.

Evaluation. The held-out script's real test split, WRR = exact match after NFC normalization, zero-width-character removal, and whitespace collapse — the

WACV #*****

[TODO: Fig. 1: pivot-space diagram — nine scripts, offset maps into shared space, grapheme-cluster vocabulary, LOSO rungs A/B/B_{bpe}]

Figure 1. The shared grapheme pivot space and the leave-one-script-out protocol. Each Brahmic script maps bijectively into a common output space by Unicode-offset correspondence; grapheme clusters in that space form the recognizer's output vocabulary. The held-out script enters training only as verified synthetic renderings (Rung B) or not at all (Rung A); evaluation is always on its real scene images.

benchmark's own protocol [8, 19] — plus character accuracy (1 − CER).

3.3. Preregistration and prospective predictions

The campaign ran under a frozen preregistration with an append-only amendment log; every amendment was filed before the results it concerns existed. Descriptors entering the transfer horse-race were computed result-blind: fertility and pivot coverage before any Rung-B result; a frozen-encoder visual-similarity control before the third. Quantitative forecasts for later runs were committed and pushed to a version-controlled archive before those runs trained (commit hashes in the supplementary material; archive anonymized for review), and are scored against a rule fixed at filing time — hits and misses alike (Sec. 5).

4. Experiments

Before any absolute number: supervised recognizers fine-tuned on real labeled target images reach ∼73% WRR on this benchmark [8]. Those systems answer a different question — they presuppose the labeled data whose absence defines our setting. The comparisons that match our question are: the source-only baseline (Rung A), the raw pre-trained VLM (WRR 0.0), the stock-BPE ablation, frontier VLMs prompted zero-shot on the same test sets, and our own forecasts.

4.1. Main result: every unseen script becomes readable

Table 1 is the campaign's core. Every one of the nine scripts moves from a near-floor source-only baseline (0.0–5.46% WRR; the raw pre-trained model reads none of them) to 9.16–30.09% WRR with character accuracy 34.57–57.77% — without a single real target image. Transfer does not collapse even on the hardest script (Malayalam, lowest pivot coverage at 79.2%: still 0.18 → 9.69). Rung-A baselines track incidental pivot-space coverage (highest for Devanagari, 5.46%), confirming the pivot shares structure before any target exposure.

		WRR (%)		Rung B	
Script	N	A	B	CharAcc	B_{bpe}
Tamil	513	0.0	9.16	39.0	2.34
Telugu	545	1.28	19.82	54.43	10.28
Kannada	720	2.78	15.42	46.37	13.33
Malayalam	547	0.18	9.69	34.57	4.2
Oriya	1044	0.77	25.38	56.37	14.94
Gujarati	1015	3.05	15.86	38.49	13.69
Bengali	2873	1.39	27.08	57.77	19.07
Devanagari	6042	5.46	30.09	57.16	27.99
Gurmukhi	2879	0.59	22.16	44.1	17.65

Table 1. Zero-real-image transfer across nine held-out Brahmic scripts (real test images). Rung A: source scripts only. Rung B: + synthetic target renderings, zero real target images. B_{bpe}: Rung B with the grapheme-cluster vocabulary removed (stock tokenizer), identical data and seed. [TODO: add per-script 95% CIs (bootstrap over samples).]

4.2. The pivot is the mechanism: BPE ablation

Removing the grapheme-cluster vocabulary — same pivot strings, same images, same seed — degrades all nine scripts (Table 1), from 1.08× (Devanagari, 30.09 → 27.99 WRR) to 3.91× (Tamil, 9.16 → 2.34): the

[TODO: Fig. 2: nine-script grouped bars (A / B / $\mathrm{B}_{\mathrm{bpe}}$), CharAcc markers overlaid]

Figure 2. Transfer by script. [TODO: generate from result JSONs (make_wacv_figs.py).]

margin varies, the direction never does (exact two-sided sign test, $p = 0.0039$). The character-accuracy pattern is diagnostic: on Telugu the BPE model matches the pivot model's character accuracy (55.1% vs. 54.43%), and on Kannada and Devanagari slightly exceeds it, yet word accuracy drops in every case. Synthetic fine-tuning alone teaches the model to see the characters; the cluster-level output space is what assembles them into correct whole words. The margin itself is not noise: it grows with the held-out script's fertility, largest exactly on the high-fertility Dravidian scripts where transfer is hardest (Sec. 5.1).

4.3. Dose–response: synthetic budget

[TODO: Preregistered Amendment 4a: malayalam/kannada/telugu $\times$ {810, 1620, 3240, 6480, 12960} synthetic images, identical recipe; per-script $\mathrm{WRR} = \alpha + \beta \log_2(\mathrm{synth})$ and joint two-factor fit; test of the frozen $c \approx +11.48$ WRR-per-doubling prediction. Fig. 3 = dose–response curves. Runs queued after the ablation completes.]

4.4. Out-of-benchmark: Khmer

[TODO: Preregistered Amendment 4b: train on all nine benchmark scripts + synthetic Khmer (different Unicode block, coeng-based stacking, outside the benchmark and outside India); evaluate on KhmerST [21]. PROSPECTIVE_PREDICTION_KHMER filed from the frozen two-factor model before training. Timeboxed; paper stands without it if the segmenter/crop engineering misses the freeze date.]

4.5. Frontier-VLM reference

[TODO: Preregistered Amendment 4c: Qwen2.5-VL-7B [2] prompted zero-shot on the same nine real test sets, same normalization and metric. Declared caveat: unknown training data makes this an upper-bound reference for off-the-shelf capability, not a like-for-like comparison. Context: frontier models read fewer than thirty of 100+ scripts [14].]

5. What Predicts Transfer?

This section reports the campaign's confirmatory analysis exactly as preregistered, including the falsification of our own primary hypothesis.

5.1. The preregistered primary fails

Fertility — the property that governs the supervised grapheme-vs-BPE trade-off (Sec. 3.1) — was preregistered as the primary transfer predictor (H3). It is refuted: Spearman $\rho = -0.80$ across the nine Rung-B results — directionally opposite to the hypothesis. The flagship case was called in advance by the declared competitor: our prospective prediction file ranked Malayalam best under fertility (P1) and worst under coverage (P2); it realized bottom-tier (9.69 WRR). We report this as a genuine falsification, consistent with fertility skepticism in the LLM-tokenizer literature [20] and with the data-centric view of STR transfer [1].

What fertility does predict (exploratory). A post-hoc analysis — not preregistered, and labeled accordingly — shows that while fertility fails to predict who transfers, it predicts who needs the pivot: across the nine scripts, the BPE-ablation penalty ratio $\mathrm{B}/\mathrm{B}_{\mathrm{bpe}}$ correlates with fertility at Spearman $\rho = +0.73$ (exact permutation $p = 0.031$). The lowest-fertility script (Devanagari) is precisely where BPE comes closest to parity (1.08×), and the highest-fertility scripts (Tamil, Malayalam) are where it collapses (3.91×, 2.31×). This mirrors, in the zero-real-image regime, the supervised relationship between fertility and grapheme advantage that motivated the pivot (Sec. 3). Two cautions: the ratio shares a denominator with absolute transfer (high-fertility scripts also transfer worst, so small denominators inflate ratios — on the absolute difference $\mathrm{B} - \mathrm{B}_{\mathrm{bpe}}$ the correlation drops to $\rho = +0.28$), and with $n = 9$ this is a candidate regularity for confirmatory testing on new scripts, not a law.

5.2. The two-factor account

Two result-blind quantities explain the campaign. (i) Synthetic budget dominates. The two scripts with the disclosed 2× synthetic budget (Bengali, Devanagari) are the top two results (27.08, 30.09 WRR) — Bengali from the second-lowest coverage. (ii) Coverage orders transfer within a budget. Among the seven equal-budget scripts, pivot-space token coverage correlates at $\rho = +0.61$ (raw, all nine: $+0.37$). The joint fit $\mathrm{WRR} = -35.79 + 0.583 \cdot \mathrm{cov} + 11.48 \cdot \mathbb{1}[2\times]$ attains RMSE 4.18 WRR across all nine scripts, with

[TODO: Fig. 4: prediction receipts — for each filed forecast, the committed point + $\pm1\sigma/\pm2\sigma$ bands vs. realized WRR, with commit-and-push timestamps relative to training start]

Figure 3. Every quantitative forecast filed during the campaign, scored against the rule fixed at filing time. Hits and misses are both shown.

both 2× residuals near zero. A frozen-encoder visual-similarity control — rendered glyphs embedded by the recognizer's own vision encoder — predicts nothing ($\rho = -0.14$), separating the structural coverage effect from the visual-similarity account of [1]. [TODO: report bootstrap CIs for both coefficients; cross-check -0.14 at 9 points.]

5.3. Prospective scorecard

Three forecasts were filed before their runs; all are scored (Fig. 3). Kannada: exploratory point ∼15 WRR, filed with two scripts observed; realized 15.42. Devanagari: directional call (high, ∼25+) committed before the result existed but pushed after — we therefore label it only weakly prospective and lean on it for nothing. Gurmukhi (final script): point 16.2 WRR, bands $\pm1\sigma$ [12.2, 20.2] / $\pm2\sigma$ [8.2, 24.1], committed and pushed 3.7 hours before training began; realized 22.16 — outside the $\pm1\sigma$ band (under-called by 1.5σ), inside $\pm2\sigma$. The declared rank bet between the two preregistered predictors resolved for coverage over fertility (ρ +0.32 vs. −0.71 on the seven forecast scripts), with neither ranking alone significant — which is why the two-factor account, not a single-predictor law, is our claim. The model's honest summary: transfer is forecastable to roughly ±4.18 WRR (1σ) from two quantities computable before training — and the misses are part of the record.

6. Limitations

Absolute accuracy. 9.16–30.09% WRR reads many words but is far below the supervised ceiling (∼73% with real labeled data [8]); the synthetic → real domain gap is the dominant residual, and our training loss reaching ∼0.06 on synthetic data while real-image WRR stays modest quantifies it. One architecture. All transfer results use Florence-2; the supervised granularity study spans 12 runs of a second architecture family, but architecture-generality of transfer is untested. One family. Nine scripts, one benchmark, one script family whose shared structure the pivot exploits by design; Khmer (Sec. 4.4) probes the boundary, and non-Brahmic generalization is open. Model imperfections. Gujarati underperforms its coverage (highest coverage, mid-table transfer), one Rung-A baseline breaches the predicted < 5 line by 0.46 (Devanagari, 5.46), and the final-script forecast under-called by 1.5σ; all are reported, none are explained away. Compute. Single 24 GB GPU; budgets (700/source, 3,240 synthetic) were fixed by preregistration, not tuned.

7. Conclusion

We showed that real scene text of an entirely unseen Brahmic script can be read — at 9.16–30.09% word accuracy across all nine scripts of a public benchmark — by a small open VLM fine-tuned in a shared grapheme pivot space with only rendered synthetic exposure to the target, and that a stock-BPE ablation collapses this transfer, isolating output-space structure as the mechanism. The campaign ran under a frozen preregistration whose primary hypothesis we falsified and report; the surviving two-factor account — synthetic budget plus pivot coverage — forecasts transfer to ∼ ± 4.18 WRR from quantities available before training, with its record of committed-in-advance hits and misses auditable by commit hash. Beyond the specific results, we offer the recipe — structural pivot, verified rendering, prospective evaluation — as a template: for the long tail of the world's scripts, predictable zero-real-image transfer may matter more than another point of accuracy on scripts that are already read.

References

[1] Jeonghun Baek, Yusuke Matsui, and Kiyoharu Aizawa. Cross-lingual learning in multilingual scene text recognition. arXiv preprint arXiv:2312.10806, 2023. 3, 5, 6

[2] Shuai Bai, Keqin Chen, Xuejing Liu, Jialin Wang, Wenbin Ge, Sibo Song, Kai Dang, Peng Wang, Shijie Wang, Jun Tang, Humen Zhong, Yuanzhi Zhu, Mingkun Yang, Zhaohai Li, Jianqiang Wan, Pengfei Wang, Wei Ding, Zheren Fu, Yiheng Xu, Jiabo Ye, Xi Zhang, Tianbao Xie, Zesen Cheng, Hang Zhang, Zhibo Yang, Haiyang Xu, and Junyang Lin. Qwen2.5-vl technical report. arXiv preprint arXiv:2502.13923, 2025. 5

[3] Darwin Bautista and Rowel Atienza. Scene text recognition with permuted autoregressive sequence models. arXiv preprint arXiv:2207.06966, 2022. 2

[4] Jingye Chen, Bin Li, and Xiangyang Xue. Zero-shot

chinese character recognition with stroke-level decomposition. arXiv preprint arXiv:2106.11613, 2021. 2

[5] Zhehuai Chen, Ankur Bapna, Andrew Rosenberg, Yu Zhang, Bhuvana Ramabhadran, Pedro Moreno, and Nanxin Chen. Maestro-u: Leveraging joint speech-text representation learning for zero supervised speech asr. arXiv preprint arXiv:2210.10027, 2022. 2

[6] Yan Hon Michael Chung and Donghyeok Choi. Fine-tuning vision-language models as ocr systems for low-resource languages: A case study of manchu. arXiv preprint arXiv:2507.06761, 2025. 2

[7] Cheng Da, Peng Wang, and Cong Yao. Multi-granularity prediction with learnable fusion for scene text recognition. arXiv preprint arXiv:2307.13244, 2023. 2

[8] Anik De, Abhirama Subramanyam Penamakuri, Rajeev Yadav, Aditya Rathore, Harshiv Shah, Devesh Sharma, Sagar Agarwal, Pravin Kumar, and Anand Mishra. Bharat scene text: A novel comprehensive dataset and benchmark for indian language scene text understanding. arXiv preprint arXiv:2511.23071, 2025. 1, 2, 3, 4, 6

[9] C. M. Downey, Terra Blevins, Dhwani Serai, Dwija Parikh, and Shane Steinert-Threlkeld. Targeted multilingual adaptation for low-resource language families. arXiv preprint arXiv:2405.12413, 2024. 3

[10] Carlos Garrido-Munoz, Aniello Panariello, Silvia Cascianelli, Angelo Porrello, Simone Calderara, Jorge Calvo-Zaragoza, and Rita Cucchiara. Zero-shot synthetic-to-real handwritten text recognition via task analogies. arXiv preprint arXiv:2604.09713, 2026. 2

[11] Md. Mahmudul Hasan, Ahmed Nesar Tahsin Choudhury, Mahmudul Hasan, and Md. Mosaddek Khan. Gradet-htr: A resource-efficient bengali handwritten text recognition system utilizing grapheme-based tokenizer and decoder-only transformer. arXiv preprint arXiv:2509.18081, 2025. 2

[12] Moritz Herrmann, F. Julian D. Lange, Katharina Eggensperger, Giuseppe Casalicchio, Marcel Wever, Matthias Feurer, David Rügamer, Eyke Hüllermeier, Anne-Laure Boulesteix, and Bernd Bischl. Position: Why we must rethink empirical research in machine learning. arXiv preprint arXiv:2405.02200, 2024. 3

[13] Jake M. Hofman, Angelos Chatzimparmpas, Amit Sharma, Duncan J. Watts, and Jessica Hullman. Preregistration for predictive modeling. arXiv preprint arXiv:2311.18807, 2023. 3

[14] Amir Hossein Kargaran, Nafiseh Nikeghbal, Jana Diesner, François Yvon, and Hinrich Schütze. Glotocr bench: Ocr models still struggle beyond a handful of unicode scripts. arXiv preprint arXiv:2604.12978, 2026. 1, 2, 5

[15] Prateek Keserwani, Taveena Lotey, Rohit Keshari, and Partha Pratim Roy. Separate scene text detector for unseen scripts is not all you need. arXiv preprint arXiv:2307.15991, 2023. 2

[16] Chang Liu, Chun Yang, Hai-Bo Qin, Xiaobin Zhu, Cheng-Lin Liu, and Xu-Cheng Yin. Towards open-set text recognition via label-to-prototype learning. arXiv preprint arXiv:2203.05179, 2022. 2

[17] Chang Liu, Chun Yang, and Xu-Cheng Yin. Open-set text recognition via character-context decoupling. arXiv preprint arXiv:2204.05535, 2022.

[18] Chang Liu, Simon Corbillé, and Elisa H Barney Smith. Moose: Multi-orientation sharing experts for open-set scene text recognition. arXiv preprint arXiv:2407.18616, 2024. 2

[19] Harsh Lunia, Ajoy Mondal, and C V Jawahar. Indicstr12: A dataset for indic scene text recognition. arXiv preprint arXiv:2403.08007, 2024. 1, 2, 4

[20] Mir Tafseer Nayeem, Sawsan Alqahtani, Md Tahmid Rahman Laskar, Tasnim Mohiuddin, and M Saiful Bari. Beyond fertility: Analyzing strr as a metric for multilingual tokenization evaluation. arXiv preprint arXiv:2510.09947, 2025. 2, 5

[21] Vannkinh Nom, Souhail Bakkali, Muhammad Muzzamil Luqman, Mickaël Coustaty, and Jean-Marc Ogier. Khmerst: A low-resource khmer scene text detection and recognition benchmark. arXiv preprint arXiv:2410.18277, 2024. 5

[22] Jirui Qi, Raquel Fernández, and Arianna Bisazza. Cross-lingual consistency of factual knowledge in multilingual language models. arXiv preprint arXiv:2310.10378, 2023. 3

[23] Miao Rang, Zhenni Bi, Chuanjian Liu, Yunhe Wang, and Kai Han. An empirical study of scaling law for ocr. arXiv preprint arXiv:2401.00028, 2023. 3

[24] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. arXiv preprint arXiv:2311.06242, 2023. 1, 3

[25] Xingsong Ye, Yongkun Du, JiaXin Zhang, Chen Li, Jing Lyu, and Zhineng Chen. What is wrong with synthetic data for scene text recognition? a strong synthetic engine with diverse simulations and self-evolution. arXiv preprint arXiv:2602.06450, 2026. 2

[26] Jinshan Zeng, Ruiying Xu, Yu Wu, Hongwei Li, and Jiaxing Lu. Star: Zero-shot chinese character recognition with stroke- and radical-level decompositions. arXiv preprint arXiv:2210.08490, 2022. 2

[27] Yuyi Zhang, Yuanzhi Zhu, Dezhi Peng, Peirong Zhang, Zhenhua Yang, Zhibo Yang, Cong Yao, and Lianwen Jin. Hiercode: A lightweight hierarchical codebook for zero-shot chinese text recognition. arXiv preprint arXiv:2403.13761, 2024. 2